

Identifying Variables — Set B

Level 3 · Ages 11-14 · Name every variable. Give values and units where you can.

Name _____ Date _____

Independent variable — the factor you change, and the values you test.

Dependent variable — the factor you measure, with its units.

Controlled variables — factors kept the same in every trial.

Control group — a separate no-treatment trial used for comparison. If there isn't one, write "none".

- ① A student drops one fizzing vitamin tablet into 200 mL of water at 10 °C, 25 °C and 40 °C, using a new tablet of the same kind each time and never stirring. She times how long each takes to stop fizzing. (Do not drink the solution.)

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

- ② Parachutes made from the same paper, with canopy areas of 100, 200, 300 and 400 cm², each carry the same paper-clip load. Each is dropped indoors from 2 m, three times, and the mean time to reach the floor is recorded.

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

- ③ A piece of pondweed in water is lit by a lamp placed 10, 20, 30 and 40 cm away. The water is kept at 25 °C. At each distance the gas bubbles (which contain oxygen) given off in one minute are counted.

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____